

Домашнее задание

Профилирование запросов в ClickHouse

Сравнение запросов с использованием первичного ключа и без него

Параметр	Значение
СУБД	ClickHouse 25.8.22
Датасет	UK Price Paid
Таблица	uk.uk_price_paid
Движок	MergeTree
Ключ сортировки / первичный ключ	ORDER BY (postcode1, postcode2, addr1, addr2)
Уровень логирования	SET send_logs_level = 'trace'

1. Цель работы

Цель работы - сравнить выполнение запросов в ClickHouse с использованием первичного ключа и без него, а также проанализировать trace-логи и вывод команды EXPLAIN indexes = 1.

2. Исходные данные и структура таблицы

Для выполнения работы была использована таблица uk.uk_price_paid с данными демонстрационного датасета UK Price Paid. Таблица создана на движке MergeTree.

```
CREATE TABLE uk.uk_price_paid
(
    price UInt32,
    date Date,
    postcode1 LowCardinality(String),
    postcode2 LowCardinality(String),
    type Enum8('terraced' = 1, 'semi-detached' = 2, 'detached' = 3, 'flat' = 4, 'other' = 0),
    is_new UInt8,
    duration Enum8('freehold' = 1, 'leasehold' = 2, 'unknown' = 0),
    addr1 String,
    addr2 String,
    street LowCardinality(String),
    locality LowCardinality(String),
    town LowCardinality(String),
    district LowCardinality(String),
    county LowCardinality(String)
)
ENGINE = MergeTree
ORDER BY (postcode1, postcode2, addr1, addr2);
```

Так как первым полем в ORDER BY является postcode1, фильтрация по этому столбцу может эффективно использовать первичный индекс MergeTree. Поле town в ключ сортировки не входит.

3. Настройки перед выполнением запросов

```
SET send_logs_level = 'trace';
SET use_uncompressed_cache = 0;
SET enable_filesystem_cache = 0;
```

Trace-логи позволяют увидеть строки Key condition, Filtering marks by primary and secondary keys, PK index has dropped ... granules, а также количество marks, выбранных для чтения.

4. Запрос без использования первичного ключа

Первый запрос фильтрует данные по полю town. Это поле не входит в ORDER BY, поэтому ClickHouse не может эффективно использовать primary key index.

```
SELECT
    count() AS deals,
    avg(price) AS avg_price
FROM uk.uk_price_paid
WHERE town = 'LONDON';
```

4.1. Результат выполнения

Показатель	Значение
deals	2 377 559

Показатель	Значение
avg_price	526 632.0572233959
Processed rows	31.35 million
Processed data	75.55 MB
Elapsed	0.091 sec

4.2. Фрагменты trace-логов

```
Key condition: unknown
Filtering marks by primary and secondary keys
PK index has dropped 0/3832 granules
Selected 10/10 parts by partition key, 10 parts by primary key,
3832/3832 marks by primary key, 3832 marks to read from 10 ranges
Read 31346259 rows, 72.05 MiB
```

Из логов видно, что условие по первичному ключу не было построено: Key condition: unknown. Индекс не отбросил гранулы: PK index has dropped 0/3832 granules. В результате были прочитаны практически все marks.

4.3. EXPLAIN для запроса без первичного ключа

```
EXPLAIN indexes = 1
SELECT
    count() AS deals,
    avg(price) AS avg_price
FROM uk.uk_price_paid
WHERE town = 'LONDON';

ReadFromMergeTree (uk.uk_price_paid)
Indexes:
  PrimaryKey
    Condition: true
    Parts: 3/3
    Granules: 3827/3827
    Ranges: 8
```

В EXPLAIN указано Condition: true и Granules: 3827/3827. Это означает, что условие town = LONDON не было использовано для ограничения чтения по primary key.

5. Запрос с использованием первичного ключа

Второй запрос фильтрует данные по полю postcode1. Это первое поле в ORDER BY, поэтому ClickHouse может использовать primary key index.

```
SELECT
    count() AS deals,
    avg(price) AS avg_price
FROM uk.uk_price_paid
WHERE postcode1 = 'SW1A';
```

5.1. Результат выполнения

Показатель	Значение
deals	537
avg_price	6 832 204.932960894
Processed rows	25.42 thousand
Processed data	92.36 KB
Elapsed	0.005 sec

5.2. Фрагменты trace-логов

```
Key condition: (column 0 in ['SW1A', 'SW1A'])
Filtering marks by primary and secondary keys
PK index has dropped 3824/3827 granules
Selected 3/3 parts by partition key, 3 parts by primary key,
3/3827 marks by primary key, 3 marks to read from 3 ranges
Read 25423 rows, 90.20 KiB
```

В логах видно, что ClickHouse построил условие по первому столбцу ключа: column 0 соответствует postcode1. Индекс отбросил 3824 гранулы из 3827, поэтому к чтению осталось только 3 гранулы.

5.3. EXPLAIN для запроса с первичным ключом

```
EXPLAIN indexes = 1
SELECT
    count() AS deals,
    avg(price) AS avg_price
FROM uk.uk_price_paid
WHERE postcode1 = 'SW1A';

ReadFromMergeTree (uk.uk_price_paid)
Indexes:
  PrimaryKey
  Keys:
    postcode1
  Condition: (postcode1 in ['SW1A', 'SW1A'])
  Parts: 3/3
  Granules: 3/3827
  Search Algorithm: binary search
  Ranges: 2
```

В EXPLAIN видно, что используется PrimaryKey, а в секции Keys указан столбец postcode1. Количество читаемых гранул уменьшилось до 3/3827.

6. Сравнение результатов

Критерий	Без PK: town = LONDON	С PK: postcode1 = SW1A
Поле входит в ORDER BY	Нет	Да, первое поле
Key condition	unknown	column 0 in ['SW1A', 'SW1A']
Отброшено гранул PK index	0/3832	3824/3827
Marks / granules к чтению	3832/3832	3/3827
Обработано строк	31.35 million	25.42 thousand
Обработано данных	75.55 MB	92.36 KB
Время выполнения	0.091 sec	0.005 sec
EXPLAIN	Condition: true	Keys: postcode1

7. Вывод

В ходе работы были выполнены два запроса к таблице uk.uk_price_paid. Первый запрос фильтровал данные по town. Это поле не входит в первичный ключ, поэтому ClickHouse не смог использовать primary key index для сокращения чтения данных. В логах было указано Key condition: unknown, а PK index has dropped 0/3832 granules. В EXPLAIN было показано Condition: true и Granules: 3827/3827.

Второй запрос фильтровал данные по postcode1. Это первое поле в ORDER BY, поэтому ClickHouse использовал primary key index. В логах было указано Key condition: (column 0 in ['SW1A', 'SW1A']) и PK index has dropped 3824/3827 granules. В EXPLAIN было показано Keys: postcode1 и Granules: 3/3827.

Таким образом, запрос с условием по первичному ключу оказался значительно эффективнее: он обработал около 25.42 тыс. строк вместо 31.35 млн строк и прочитал 92.36 KB вместо 75.55 MB.

Приложение. Скриншоты выполнения запросов

Скриншот 1. Запрос без первичного ключа и trace-логи

```
compute-vm-2-4-30-ssd-1776951461260.ru-central1.internal :) SELECT
  count() AS deals,
  avg(price) AS avg_price
FROM uk.uk_price_paid
WHERE town = 'LONDON';

SELECT
  count() AS deals,
  avg(price) AS avg_price
FROM uk.uk_price_paid
WHERE town = 'LONDON'
```

Query id: 2893e6a9-1001-49fc-ad36-c7bf12bd741d

```
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.050683 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Debug> executeQuery: (from 127.0.0.1:45816) (query 1, line
1) SELECT count() AS deals, avg(price) AS avg_price FROM uk.uk_price_paid WHERE town = 'LONDON'; (stage: Complete)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.051512 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> Planner: Query to stage Complete
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.051698 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> Planner: Query from stage FetchColumns to stage Com
plete
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.052077 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> Aggregator: Adjusting memory limit before external
aggregation with 1.51 GiB (ratio: 0.5, available system memory: 3.02 GiB)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.052391 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> QueryPlanOptimizePrewrite: The min valid primary ke
y position for moving to the tail of Prewrite is -1
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.052460 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> QueryPlanOptimizePrewrite: Moved 1 conditions to PR
EWRITE
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.052611 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> InterpreterUnionOrSelectQuery: The new analyzer is
enabled, but the old interpreter is used. It can be a bug, please report it. Will disable 'allow_experimental_analyzer' setting (for query: SELECT min(postcode1), max(post
code1), count()) SETTINGS aggregate_functions_null_for_empty = false, transform_null_in = false, legacy_column_name_of_tuple_literal = false)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.052948 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd
069) (SelectExecutor): Key condition: unknown
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.053049 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd
069) (SelectExecutor): Filtering marks by primary and secondary keys
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.053813 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd
069) (SelectExecutor): PK index has dropped 0/3832 granules, it took 0ms across 2 threads.
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.053853 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd
069) (SelectExecutor): Query condition cache has dropped 0/3832 granules for Prewrite condition equals(__table1.town, 'LONDON'.String).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.053976 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd
069) (SelectExecutor): Query condition cache has dropped 0/3832 granules for WHERE condition equals(town, 'LONDON'.String).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.053898 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd
069) (SelectExecutor): Selected 10/10 parts by partition key, 10 parts by primary key, 3832/3832 marks by primary key, 3832 marks to read from 10 ranges
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.053910 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd
069) (SelectExecutor): Spreading mark ranges among streams (default reading)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.054090 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd
069) (SelectExecutor): Reading approx. 31346259 rows with 2 streams
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.055510 [ 2567 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> AggregatingTransform: Aggregating
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.055538 [ 2567 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> Aggregator: Aggregation method: without_key
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.057055 [ 2539 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> AggregatingTransform: Aggregating
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.057082 [ 2539 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> Aggregator: Aggregation method: without_key
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.139885 [ 2567 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> AggregatingTransform: Aggregated. 909745 to 1 rows
(from 3.47 MiB) in 0.08564585 sec. (10622172.586 rows/sec., 40.52 MiB/sec.)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.140010 [ 2539 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> AggregatingTransform: Aggregated. 1467814 to 1 rows
(from 5.60 MiB) in 0.085780232 sec. (17111331.664 rows/sec., 65.27 MiB/sec.)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.140033 [ 2539 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> Aggregator: Merging aggregated data
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.140049 [ 2539 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Trace> HashTablesStatistics: Statistics updated for key=67
19678956845330: sum_of_sizes=2, median_size=1
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.140932 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Debug> executeQuery: Read 31346259 rows, 72.05 MiB in 0.09
0293 sec., 347161562.9118536 rows/sec., 797.98 MiB/sec.
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:41:46.140963 [ 1166 ] {2893e6a9-1001-49fc-ad36-c7bf12bd741d} <Debug> MemoryTracker: Query peak memory usage: 998.81 KiB.
```

deals	avg_price
2377559	526632.0572233959

1 row in set. Elapsed: 0.091 sec. Processed 31.35 million rows, 75.55 MB (345.98 million rows/s., 833.90 MB/s.)
Peak memory usage: 998.81 KiB.

Скриншот 2. EXPLAIN для запроса без первичного ключа

```
compute-vm-2-4-30-ssd-1776951461260.ru-central1.internal :) EXPLAIN indexes = 1
SELECT
```

```
    count() AS deals,
    avg(price) AS avg_price
FROM uk.uk_price_paid
WHERE town = 'LONDON';
```

```
EXPLAIN indexes = 1
```

```
SELECT
    count() AS deals,
    avg(price) AS avg_price
FROM uk.uk_price_paid
WHERE town = 'LONDON'
```

Query id: 5dce785e-e2a3-4440-8a26-d816c849d50e

```
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:42:52.134562 [ 1166 ] {5dce785e-e2a3-4440-8a26-d816c849d50e} <Debug> executeQuery: (from 127.0.0.1:45816) (query 1, line
1) EXPLAIN indexes = 1 SELECT count() AS deals, avg(price) AS avg_price FROM uk.uk_price_paid WHERE town = 'LONDON'; (stage: Complete)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:42:52.135081 [ 1166 ] {5dce785e-e2a3-4440-8a26-d816c849d50e} <Trace> Planner: Query to stage Complete
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:42:52.135294 [ 1166 ] {5dce785e-e2a3-4440-8a26-d816c849d50e} <Trace> Planner: Query from stage FetchColumns to stage Com
plete
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:42:52.135448 [ 1166 ] {5dce785e-e2a3-4440-8a26-d816c849d50e} <Trace> Aggregator: Adjusting memory limit before external
aggregation with 1.54 GiB (ratio: 0.5, available system memory: 3.09 GiB)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:42:52.135664 [ 1166 ] {5dce785e-e2a3-4440-8a26-d816c849d50e} <Trace> QueryPlanOptimizePrewrite: The min valid primary ke
y position for moving to the tail of Prewrite is -1
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:42:52.135684 [ 1166 ] {5dce785e-e2a3-4440-8a26-d816c849d50e} <Trace> QueryPlanOptimizePrewrite: Moved 1 conditions to PR
EWRITE
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:42:52.135819 [ 1166 ] {5dce785e-e2a3-4440-8a26-d816c849d50e} <Trace> IInterpreterUnionOrSelectQuery: The new analyzer is
enabled, but the old interpreter is used. It can be a bug, please report it. Will disable 'allow_experimental_analyzer' setting (for query: SELECT min(postcode1), max(post
code1), count()) SETTINGS aggregate_functions_null_for_empty = false, transform_null_in = false, legacy_column_name_of_tuple_literal = false)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:42:52.136128 [ 1166 ] {5dce785e-e2a3-4440-8a26-d816c849d50e} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd
069) (SelectExecutor): Key condition: unknown
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:42:52.136202 [ 1166 ] {5dce785e-e2a3-4440-8a26-d816c849d50e} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd
069) (SelectExecutor): Filtering marks by primary and secondary keys
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:42:52.136595 [ 1166 ] {5dce785e-e2a3-4440-8a26-d816c849d50e} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd
069) (SelectExecutor): PK index has dropped 0/3827 granules, it took 0ms across 2 threads.
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:42:52.136631 [ 1166 ] {5dce785e-e2a3-4440-8a26-d816c849d50e} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd
069) (SelectExecutor): Query condition cache has dropped 69/3827 granules for Prewrite condition equals(__table1.town, 'LONDON'_String).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:42:52.136647 [ 1166 ] {5dce785e-e2a3-4440-8a26-d816c849d50e} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd
069) (SelectExecutor): Query condition cache has dropped 0/3758 granules for WHERE condition equals(town, 'LONDON'_String).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:42:52.137313 [ 1166 ] {5dce785e-e2a3-4440-8a26-d816c849d50e} <Debug> executeQuery: Read 11 rows, 425.00 B in 0.002817 se
c., 3904.8633297834576 rows/sec., 147.33 KiB/sec.
```

```
explain
1. Expression ((Project names + Projection))
2.   Aggregating
3.     Expression (Before GROUP BY)
4.       Expression ((WHERE + Change column names to column identifiers))
5.         ReadFromMergeTree (uk.uk_price_paid)
6.         Indexes:
7.           PrimaryKey
8.             Condition: true
9.             Parts: 3/3
10.            Granules: 3827/3827
11.            Ranges: 8
```

11 rows in set. Elapsed: 0.003 sec.

Скриншот 3. Запрос с первичным ключом и трассе-логи

```
compute-vm-2-4-30-ssd-1776951461260.ru-central1.internal :) SELECT
  count() AS deals,
  avg(price) AS avg_price
FROM uk.uk_price_paid
WHERE postcode1 = 'SW1A';
```

```
SELECT
  count() AS deals,
  avg(price) AS avg_price
FROM uk.uk_price_paid
WHERE postcode1 = 'SW1A'
```

Query id: 14a33053-003e-40fe-8e9d-58b8df8eb091

```
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.721010 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> executeQuery: (from 127.0.0.1:45816) (query 1, line 1) SELECT count() AS deals, avg(price) AS avg_price FROM uk.uk_price_paid WHERE postcode1 = 'SW1A'; (stage: Complete)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.721528 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> Planner: Query to stage Complete
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.721644 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> Planner: Query from stage FetchColumns to stage Complete
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.721768 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> Aggregator: Adjusting memory limit before external aggregation with 1.54 GiB (ratio: 0.5, available system memory: 3.07 GiB)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.721942 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> QueryPlanOptimizePrewrite: The min valid primary key position for moving to the tail of PREWHERE is 0
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.721965 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> QueryPlanOptimizePrewrite: Moved 1 conditions to PREWHERE
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722096 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> IInterpreterUnionOrSelectQuery: The new analyzer is enabled, but the old interpreter is used. It can be a bug, please report it. Will disable 'allow_experimental_analyzer' setting (for query: SELECT min(postcode1), max(postcode1), count() SETTINGS aggregate_functions_null_for_empty = false, transform_null_in = false, legacy_column_name_of_tuple_literal = false)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722451 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Key condition : (column 0 in ['SW1A', 'SW1A'])
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722493 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Filtering marks by primary and secondary keys
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722636 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binary search on index range for part all_1_28_2 (3691 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722706 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT) boundary mark: 3176
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722749 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT) boundary mark: 3177
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722776 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continuous range in 21 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722913 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binary search on index range for part all_29_29_0 (134 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722983 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT) boundary mark: 114
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723014 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT) boundary mark: 115
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723039 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continuous range in 12 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723130 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binary search on index range for part all_30_30_0 (5 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723161 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT) boundary mark: 3
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723184 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT) boundary mark: 4
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723211 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continuous range in 2 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723417 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): PK index has dropped 3824/3827 granules, it took 0ms across 2 threads.
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723468 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Query condition on cache has dropped 0/3 granules for PREWHERE condition equals(_table1.postcode1, 'SW1A' String).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723483 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Query condition on cache has dropped 0/3 granules for WHERE condition equals(postcode1, 'SW1A' String).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723505 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Selected 3/3 parts by partition key, 3 parts by primary key, 3/3827 marks by primary key, 3 marks to read from 3 ranges
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723527 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Spreading mark ranges among streams (default reading)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723640 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Reading approximately 25423 rows with 2 streams
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.724764 [ 2661 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> AggregatingTransform: Aggregating
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.724790 [ 2661 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> Aggregator: Aggregation method: without_key
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.725065 [ 2661 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> AggregatingTransform: Aggregated. 17 to 1 rows (from 68.00 B) in 0.00129916 sec. (13085.378 rows/sec., 51.11 KIB/sec.)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.725344 [ 2580 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> AggregatingTransform: Aggregating
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.725360 [ 2580 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> Aggregator: Aggregation method: without_key
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.725427 [ 2580 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> AggregatingTransform: Aggregated. 520 to 1 rows (from 2.03 KIB) in 0.001663777 sec. (312541.885 rows/sec., 1.19 MiB/sec.)
```


Скриншот 4. Детализация логов запроса с первичным ключом

```
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.721965 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> QueryPlanOptimizePrewrite: Moved 1 conditions to PREWHERE
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722096 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> IInterpreterUnionOrSelectQuery: The new analyzer is enabled, but the old interpreter is
used. It can be a bug, please report it. Will disable 'allow_experimental_analyzer' setting (for query: SELECT min(postcode1), max(postcode1), count()) SETTINGS aggregate_functions_null_for_empty = false, tra
nsform_null_in = false, legacy_column_name_of_tuple_literal = false)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722451 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Key condition
: (column 0 in ['SW1A', 'SW1A'])
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722493 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Filtering mar
ks by primary and secondary keys
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722636 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binar
y search on index range for part all_1_28_2 (3691 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722706 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT)
boundary mark: 3176
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722749 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT)
boundary mark: 3177
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722776 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continu
ous range in 21 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722913 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binar
y search on index range for part all_29_20_0 (134 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.722983 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT)
boundary mark: 114
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723014 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT)
boundary mark: 115
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723039 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continu
ous range in 12 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723130 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binar
y search on index range for part all_30_30_0 (5 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723161 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT)
boundary mark: 3
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723184 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT)
boundary mark: 4
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723211 [ 2541 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continu
ous range in 2 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723417 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): PK index has
dropped 3824/3827 granules, it took 0ms across 2 threads.
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723468 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Query conditi
on cache has dropped 0/3 granules for PREWHERE condition equals(__table1.postcode1, 'SW1A'.String).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723483 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Query conditi
on cache has dropped 0/3 granules for WHERE condition equals(postcode1, 'SW1A'.String).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723505 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Selected 3/3
parts by partition key, 3 parts by primary key, 3/3827 marks by primary key, 3 marks to read from 3 ranges
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723527 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Spreading mar
k ranges among streams (default reading)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.723640 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Reading appro
x. 25423 rows with 2 streams
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.724764 [ 2661 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> AggregatingTransform: Aggregating
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.724790 [ 2661 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> Aggregator: Aggregation method: without_key
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.725065 [ 2661 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> AggregatingTransform: Aggregated. 17 to 1 rows (from 68.00 B) in 0.00129916 sec. (13085
.378 rows/sec., 51.11 KiB/sec.)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.725344 [ 2580 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> AggregatingTransform: Aggregating
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.725360 [ 2580 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> Aggregator: Aggregation method: without_key
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.725427 [ 2580 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> AggregatingTransform: Aggregated. 520 to 1 rows (from 2.03 KiB) in 0.001663777 sec. (31
2541.885 rows/sec., 1.19 MiB/sec.)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.725438 [ 2580 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> Aggregator: Merging aggregated data
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.725453 [ 2580 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Trace> HashTablesStatistics: Statistics updated for key=6875533511731881691: sum_of_sizes=2, m
edian_size=1
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.726277 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> executeQuery: Read 25423 rows, 90.20 KiB in 0.005302 sec., 4794983.025273481 rows/sec.,
16.61 MiB/sec.
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:43:28.726299 [ 1166 ] {14a33053-003e-40fe-8e9d-58b8df8eb091} <Debug> MemoryTracker: Query peak memory usage: 673.26 KiB.
```

	deals	avg_price
1.	537	6832204.932960894

1 row in set. Elapsed: 0.005 sec. Processed 25.42 thousand rows, 92.36 KB (4.85 million rows/s., 17.64 MB/s.)
Peak memory usage: 673.26 KiB.

Скриншот 5. EXPLAIN для запроса с первичным ключом

```
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.484549 [ 1166 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Debug> executeQuery: (from 127.0.0.1:45816) (query 1, line 1) EXPLAIN indexes = 1 SELECT count
() AS deals, avg(price) AS avg_price FROM uk.uk_price_paid WHERE postcode1 = 'SW1A'; (stage: Complete)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.485191 [ 1166 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> Planner: Query to stage Complete
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.485379 [ 1166 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> Planner: Query from stage FetchColumns to stage Complete
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.485529 [ 1166 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> Aggregator: Adjusting memory limit before external aggregation with 1.51 GiB (ratio: 0.
5, available system memory: 3.03 GiB)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.485730 [ 1166 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> QueryPlanOptimizePrewrite: The min valid primary key position for moving to the tail of
PREWHERE is 0
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.485752 [ 1166 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> QueryPlanOptimizePrewrite: Moved 1 conditions to PREWHERE
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.485879 [ 1166 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> IInterpreterUnionOrSelectQuery: The new analyzer is enabled, but the old interpreter is
used. It can be a bug, please report it. Will disable 'allow_experimental_analyzer' setting (for query: SELECT min(postcode1), max(postcode1), count() SETTINGS aggregate_functions_null_for_empty = false, tra
nsform_null_in = false, legacy_column_name_of_tuple_literal = false)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486203 [ 1166 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Key condition
: (column 0 in ['SW1A', 'SW1A'])
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486246 [ 1166 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Filtering mar
ks by primary and secondary keys
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486393 [ 2585 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binar
y search on index range for part all_29_29_0 (134 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486393 [ 2558 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binar
y search on index range for part all_1_28_2 (3691 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486459 [ 2585 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT)
boundary mark: 114
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486467 [ 2558 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT)
boundary mark: 3176
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486488 [ 2585 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT)
boundary mark: 115
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486503 [ 2558 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT)
boundary mark: 3177
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486509 [ 2585 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continu
ous range in 12 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486517 [ 2558 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continu
ous range in 21 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486631 [ 2585 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binar
y search on index range for part all_30_30_0 (5 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486667 [ 2585 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT)
boundary mark: 3
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486690 [ 2585 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT)
boundary mark: 4
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486713 [ 2585 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continu
ous range in 2 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486872 [ 1166 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): PK index has
dropped 3824/3827 granules, it took 0ms across 2 threads.
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486932 [ 1166 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Query conditi
on cache has dropped 1/3 granules for PREWHERE condition equals(_table1.postcode1, 'SW1A'_String).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.486959 [ 1166 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Query conditi
on cache has dropped 0/2 granules for WHERE condition equals(postcode1, 'SW1A'_String).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:10.487665 [ 1166 ] {99bf43a6-35c0-4ee8-9df5-e7c7b7546dce} <Debug> executeQuery: Read 14 rows, 556.00 B in 0.003181 sec., 4401.131719585036 rows/sec., 170
.69 KiB/sec.
```

```
explain
1. Expression ((Project names + Projection))
2.   Aggregating
3.     Expression (Before GROUP BY)
4.       Expression ((WHERE + Change column names to column identifiers))
5.         ReadFromMergeTree (uk.uk_price_paid)
6.         Indexes:
7.           PrimaryKey
8.             Keys:
9.               postcode1
10.            Condition: (postcode1 in ['SW1A', 'SW1A'])
11.            Parts: 3/3
12.            Granules: 3/3827
13.            Search Algorithm: binary search
14.            Ranges: 2
```


Скриншот 6. Дополнительный запрос по двум полям ключа

```
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.773036 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Debug> executeQuery: (from 127.0.0.1:45816) (query 1, line 1) SELECT count() AS deals, avg(price) AS avg_price FROM uk.uk_price_paid WHERE postcode1 = 'SW1A' AND postcode2 = '1AA'; (stage: Complete)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.773620 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> Planner: Query to stage Complete
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.773743 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> Planner: Query from stage FetchColumns to stage Complete
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.773909 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> Aggregator: Adjusting memory limit before external aggregation with 1.51 GiB (ratio: 0.5, available system memory: 3.02 GiB)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.774110 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> QueryPlanOptimizePrewrite: The min valid primary key position for moving to the tail of PREWHERE is 1
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.774138 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> QueryPlanOptimizePrewrite: Moved 2 conditions to PREWHERE
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.774489 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> InterpreterUnionOrSelectQuery: The new analyzer is enabled, but the old interpreter is used. It can be a bug, please report it. Will disable 'allow_experimental_analyzer' setting (for query: SELECT min(postcode1), max(postcode1), count() SETTINGS aggregate_functions_null_for_empty = false, transform_null_in = false, legacy_column_name_of_tuple_literal = false)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.774803 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Key condition: (column 0 in ['SW1A', 'SW1A']), (column 1 in ['1AA', '1AA']), and
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.774902 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Filtering marks by primary and secondary keys
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775069 [ 2542 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binary search on index range for part all_1_28_2 (3691 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775118 [ 2579 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binary search on index range for part all_29_29_0 (134 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775141 [ 2542 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT) boundary mark: 3176
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775188 [ 2542 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT) boundary mark: 3177
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775209 [ 2542 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continuous range in 21 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775241 [ 2579 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT) boundary mark: 114
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775361 [ 2579 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT) boundary mark: 115
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775439 [ 2579 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continuous range in 12 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775564 [ 2579 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binary search on index range for part all_30_30_0 (5 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775632 [ 2579 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT) boundary mark: 3
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775694 [ 2579 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT) boundary mark: 4
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775764 [ 2579 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continuous range in 2 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775934 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): PK index has dropped 3824/3827 granules, it took 0ms across 2 threads.
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.775990 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Query condition on cache has dropped 0/3 granules for PREWHERE condition and(equals(_table1.postcode2, '1AA'_String), equals(_table1.postcode1, 'SW1A'_String)).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.776017 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Query condition on cache has dropped 0/3 granules for WHERE condition and(equals(postcode1, 'SW1A'_String), equals(postcode2, '1AA'_String)).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.776043 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Selected 3/3 parts by partition key, 3 parts by primary key, 3/3827 marks by primary key, 3 marks to read from 3 ranges
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.776054 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Spreading marks ranges among streams (default reading)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.776214 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Reading approximately 25423 rows with 2 streams
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.777856 [ 2551 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> Aggregator: Aggregation method: without_key
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.777888 [ 2551 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> AggregatingTransform: Aggregated. 0 to 1 rows (from 0.00 B) in 0.001486621 sec. (0.000 rows/sec., 0.00 B/sec.)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.777958 [ 2581 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> Aggregator: Aggregation method: without_key
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.777998 [ 2581 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> AggregatingTransform: Aggregated. 0 to 1 rows (from 0.00 B) in 0.001599561 sec. (0.000 rows/sec., 0.00 B/sec.)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.778004 [ 2581 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> Aggregator: Merging aggregated data
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.778019 [ 2581 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Trace> HashTablesStatistics: Statistics updated for key=1190804967503449077: sum_of_sizes=2, median_size=1
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.778692 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Debug> executeQuery: Read 25423 rows, 123.20 KiB in 0.005722 sec., 4443026.91366655 rows/sec., 21.03 MiB/sec.
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:45:34.778712 [ 1166 ] {c358a852-8e44-4d52-ad7e-55885191f993} <Debug> MemoryTracker: Query peak memory usage: 338.54 KiB.
```

	deals	avg_price
1.	0	nan

Скриншот 7. EXPLAIN для дополнительного запроса

```
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.763579 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Debug> executeQuery: (from 127.0.0.1:45816) (query 1, line 1) EXPLAIN indexes = 1 SELECT count
() AS deals, avg(price) AS avg_price FROM uk.uk_price_paid WHERE postcode1 = 'SW1A' AND postcode2 = '1AA'; (stage: Complete)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.764228 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> Planner: Query to stage Complete
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.764374 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> Planner: Query from stage FetchColumns to stage Complete
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.764540 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> Aggregator: Adjusting memory limit before external aggregation with 1.50 GiB (ratio: 0.
5, available system memory: 3.01 GiB)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.764731 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> QueryPlanOptimizePrewhere: The min valid primary key position for moving to the tail of
PREWHERE is 1
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.764754 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> QueryPlanOptimizePrewhere: Moved 2 conditions to PREWHERE
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.764902 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> InterpreterUnionOrSelectQuery: The new analyzer is enabled, but the old interpreter is
used. It can be a bug, please report it. Will disable 'allow_experimental_analyzer' setting (for query: SELECT min(postcode1), max(postcode1), count() SETTINGS aggregate_functions_null_for_empty = false, tra
nsform_null_in = false, legacy_column_name_of_tuple_literal = false)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765200 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Key condition
: (column 0 in ['SW1A', 'SW1A']), (column 1 in ['1AA', '1AA']), and
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765287 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Filtering mar
ks by primary and secondary keys
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765435 [ 2536 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binar
y search on index range for part all_1_28_2 (3691 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765436 [ 2554 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binar
y search on index range for part all_29_29_0 (134 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765498 [ 2554 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT)
boundary mark: 114
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765512 [ 2536 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT)
boundary mark: 3176
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765521 [ 2554 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT)
boundary mark: 115
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765541 [ 2554 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continu
ous range in 12 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765550 [ 2536 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT)
boundary mark: 3177
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765562 [ 2536 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continu
ous range in 21 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765638 [ 2536 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Running binar
y search on index range for part all_30_30_0 (5 marks)
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765694 [ 2536 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (LEFT)
boundary mark: 3
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765740 [ 2536 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found (RIGHT)
boundary mark: 4
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765805 [ 2536 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Trace> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Found continu
ous range in 2 steps
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.765980 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): PK index has
dropped 3824/3827 granules, it took 0ms across 2 threads.
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.766032 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Query conditi
on cache has dropped 3/3 granules for PREWHERE condition and(equals(__table1.postcode2, '1AA'_String), equals(__table1.postcode1, 'SW1A'_String)).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.766052 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Debug> uk.uk_price_paid (5edca6c4-a54d-4c72-8108-8d0b5b9cd069) (SelectExecutor): Query conditi
on cache has dropped 0/0 granules for WHERE condition and(equals(postcode1, 'SW1A'_String), equals(postcode2, '1AA'_String)).
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.766734 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Debug> executeQuery: Read 15 rows, 623.00 B in 0.00325 sec., 4615.384615384615 rows/sec., 187.
20 KiB/sec.
[compute-vm-2-4-30-ssd-1776951461260] 2026.07.01 07:46:14.766755 [ 1166 ] {600fb230-487b-4dbe-94f7-9ddbe7cf594a} <Debug> MemoryTracker: Query peak memory usage: 139.51 KiB.
```

```
explain
1. Expression ((Project names + Projection))
2. Aggregating
3. Expression (Before GROUP BY)
4. Expression ((WHERE + Change column names to column identifiers))
5. ReadFromMergeTree (uk.uk_price_paid)
6. Indexes:
7. PrimaryKey
8. Keys:
9. postcode1
10. postcode2
11. Condition: and((postcode2 in ['1AA', '1AA']), (postcode1 in ['SW1A', 'SW1A']))
12. Parts: 3/3
13. Granules: 3/3827
14. Search Algorithm: binary search
15. Ranges: 0
```

15 rows in set. Elapsed: 0.004 sec.